GPS Spoofing Demonstration Using HackRF One & DragonOS

Aviation Security Research Report

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1. Introduction

Global Navigation Satellite Systems (GNSS), especially GPS, form a critical component of modern aviation. Aircraft depend heavily on GPS for navigation, positioning, and timing. However, in recent years, GPS spoofing attacks have increased globally, affecting navigation systems of both civilian and commercial aircraft.

This report presents a controlled, demonstration of how GPS spoofing can work using a Software Defined Radio (SDR) platform — specifically HackRF One and GPS-SDR-SIM — to simulate the effect of a spoofed GPS signal near Indira Gandhi International (IGI) Airport, Delhi.

⚠️ Disclaimer:
This experiment was conducted *strictly in an isolated, non-radiating test environment*.
The intent is to raise awareness of aviation GNSS vulnerabilities for cybersecurity research and education.

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2. Tools & Technologies Used

- DragonOS (Linux-based SDR distribution)
- HackRF One with Mayhem Firmware
- GPS-SDR-SIM (GPS signal simulator)
- NASA CDDIS Ephemeris (BRDC) files
- GCC Compiler
- GPS Antenna (Test Environment Only)
- HackRF host utilities (hackrf_transfer)

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3. Objective of the Demonstration

The objective of this research is to:

- Demonstrate how a spoofed GPS signal can be generated using openly available data.
- Show the risk spoofed GPS signals pose to aviation systems dependent on GNSS.
- Highlight the importance of implementing GNSS anti-spoofing and authentication techniques.

The demonstration simulates a GPS signal that appears to come from the coordinates of IGI Airport, New Delhi.

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4. Installing & Preparing the Environment

4.1 DragonOS

DragonOS is used due to its pre-installed SDR tools, libraries, and compatibility with HackRF.

Boot into DragonOS and ensure basic SDR utilities are working.

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5. Downloading & Compiling GPS-SDR-SIM

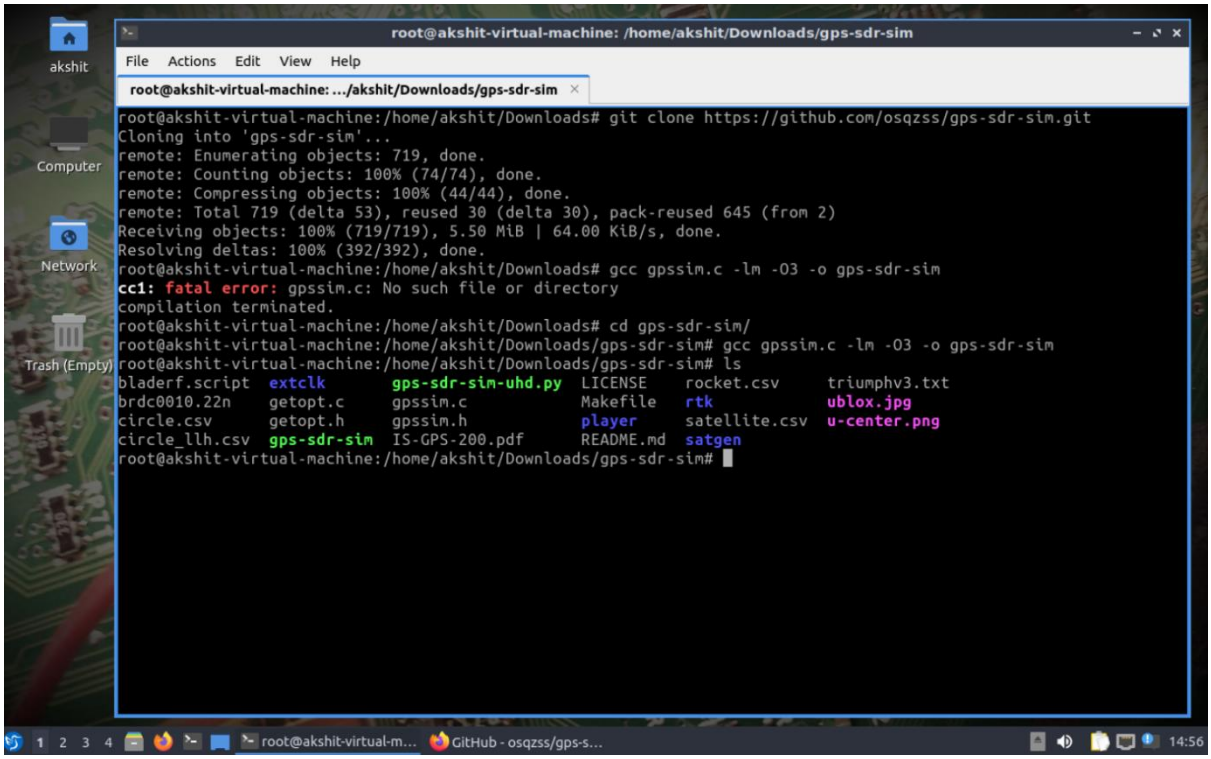
5.1 Clone GPS-SDR-SIM

“git clone https://github.com/osqzss/gps-sdr-sim.git”

5.2 Compile the C Source Code

Navigate into the cloned directory and run:

“gcc gpssim.c -lm -O3 -o gps-sdr-sim”



This generates the executable gps-sdr-sim.

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6. Downloading GPS Ephemeris Data (BRDC File)

6.1 What is an Ephemeris File?

NASA maintains daily GPS satellite broadcast navigation data known as Ephemeris (RINEX Navigation Files).
These provide satellite:

- Position
- Time
- Pseudo-range
- Doppler shifts

GPS-SDR-SIM uses this data to generate realistic spoofing waveforms.

6.2 Downloading the Latest File

1. Create/login to your Earthdata account: https://urs.earthdata.nasa.gov/login
2. Navigate to the daily GNSS archive: https://cddis.nasa.gov/archive/gnss/data/daily/

Example file used:

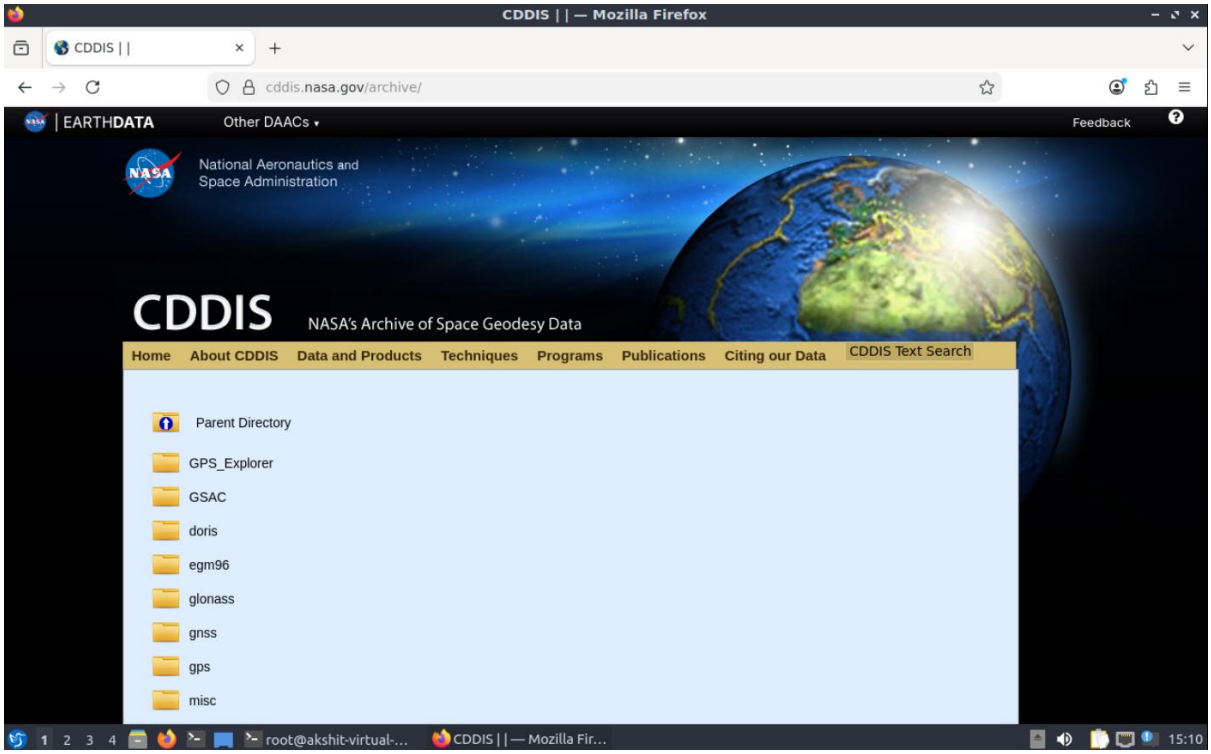
brdc3190.25n.gz

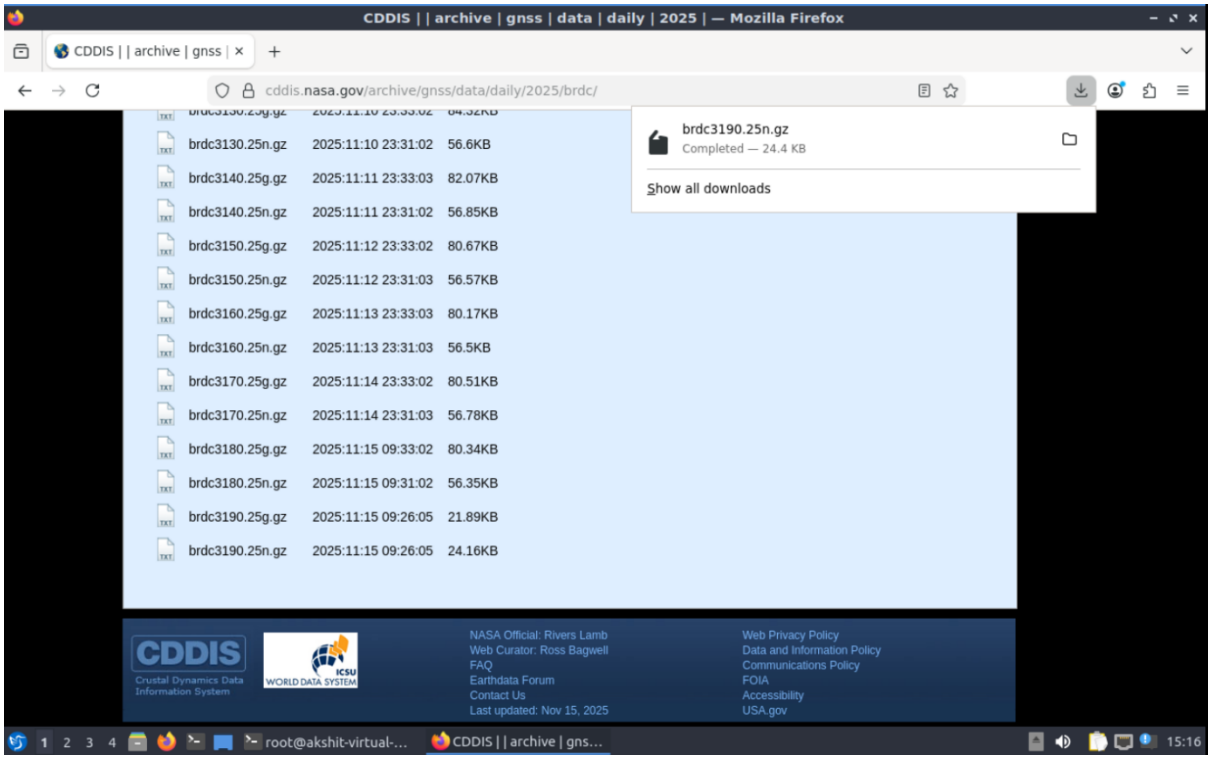
6.3 Extract the File

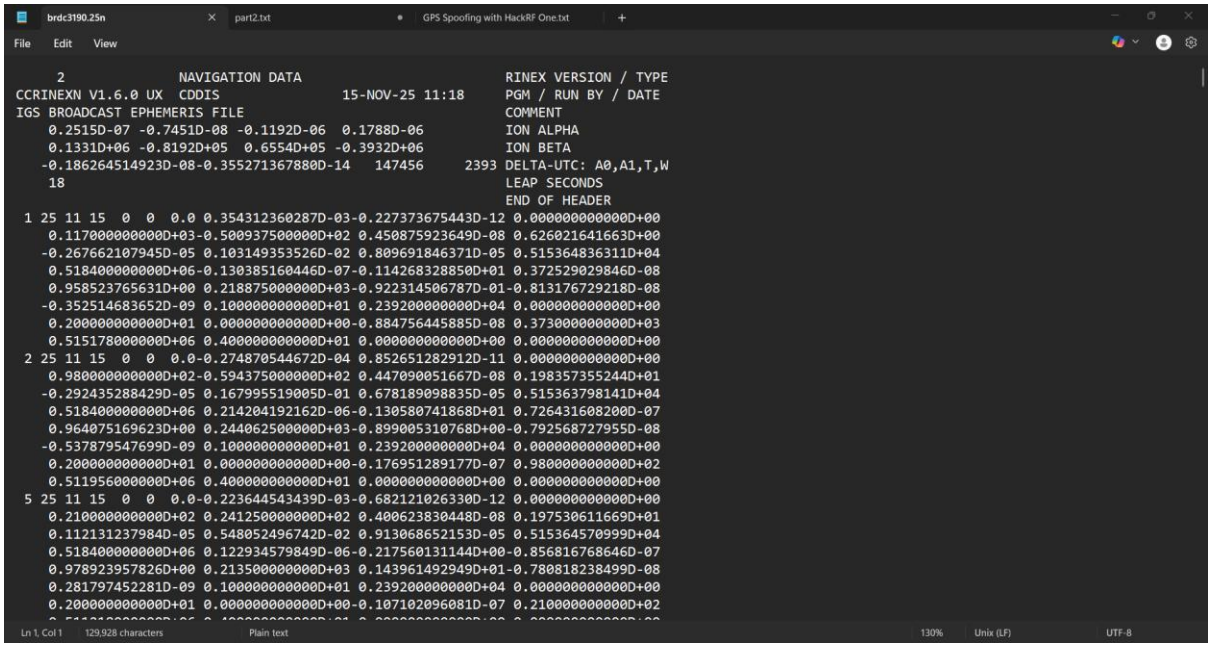
gzip -d brdc3190.25n.gz

This extracts:

brdc3190.25n


CDDIS | | — Mozilla Firefox
CDDIS | |
cddis.nasa.gov/archive/
EARTHDATA
Other DAACs
Feedback
National Aeronautics and Space Administration
CDDIS
NASA's Archive of Space Geodesy Data
Home
About CDDIS
Data and Products
Techniques
Programs
Publications
Citing our Data
CDDIS Text Search
Parent Directory
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CDDIS
Crustal Dynamics Data Information System
ICSU WORLD DATA SYSTEM
NASA Official: Rivers Lamb
Web Curator: Ross Bagwell
FAQ
Earthdata Forum
Contact Us
Last updated: Nov 15, 2025
Web Privacy Policy
Data and Information Policy
Communications Policy
FOIA
Accessibility
USA.gov
root@akshit-virtual-...
CDDIS | | archive | gns...
15:16



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7. Generating the Spoofed GPS Signal

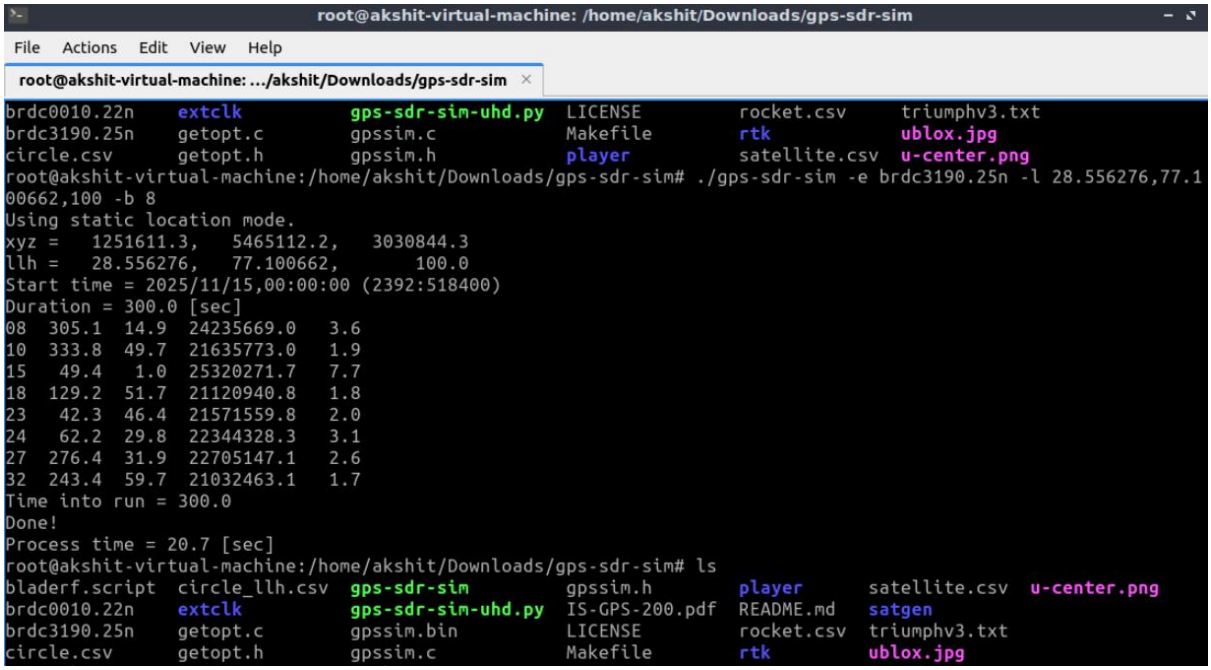
We generate the I/Q samples needed for HackRF transmission.

7.1 Command to Generate gpssim.bin

“./gps-sdr-sim -e brdc3190.25n -l 28.556276,77.100662,100 -b 8”

Where:

- 28.556276 = Latitude of IGI Airport
- 77.100662 = Longitude of IGI Airport
- 100 meters = Altitude
- -b 8 = 8-bit I/Q samples required for HackRF



This produces:

gpssim.bin

This file contains the spoofed GPS simulation waveform for ~300 seconds (default duration).

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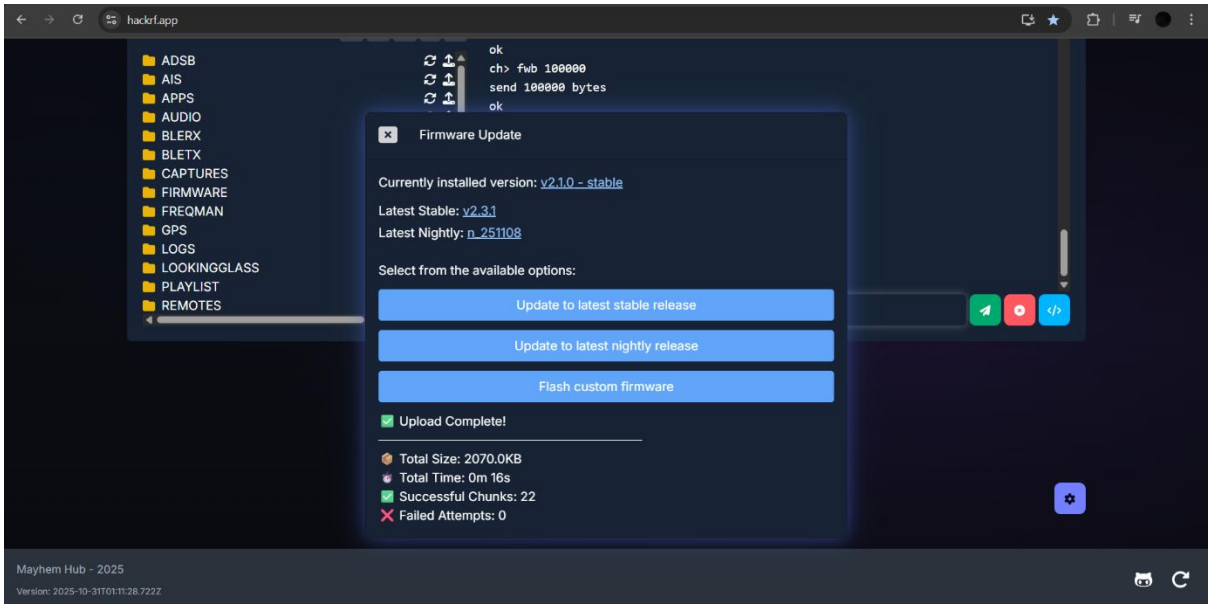
8. Preparing HackRF One

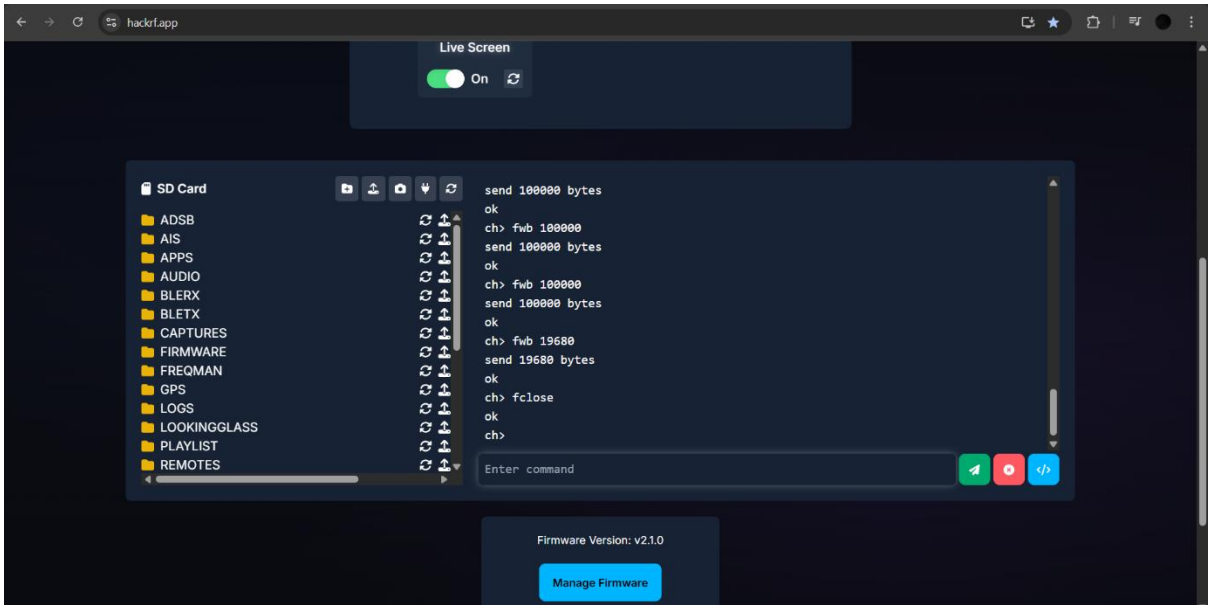
8.1 Updating Firmware

Mayhem firmware updated via:

https://hackrf.app/







8.2 Verifying HackRF

hackrf_info

```
root@akshit-virtual-machine:/home/akshit/Downloads/gps-sdr-sim# lsusb
Bus 002 Device 003: ID 1d50:6089 OpenMoko, Inc. Great Scott Gadgets HackRF One SDR
Bus 002 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 003: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 001 Device 002: ID 0e0f:0003 VMware, Inc. Virtual Mouse
Bus 001 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
root@akshit-virtual-machine:/home/akshit/Downloads/gps-sdr-sim# hackrf_info
hackrf_info version: 2024.02.1
libhackrf version: 2024.02.1 (0.9)
Found HackRF
Index: 0
Serial number: 0000000000000000042a068dc34a54f07
Board ID Number: 4 (HackRF One)
Firmware Version: v2.3.1 (API:1.08)
Part ID Number: 0xa000cb3c 0x00614f66
Hardware Revision: r9
Hardware does not appear to have been manufactured by Great Scott Gadgets.
Hardware supported by installed firmware:
    HackRF One
root@akshit-virtual-machine:/home/akshit/Downloads/gps-sdr-sim#
```

Device should be detected.

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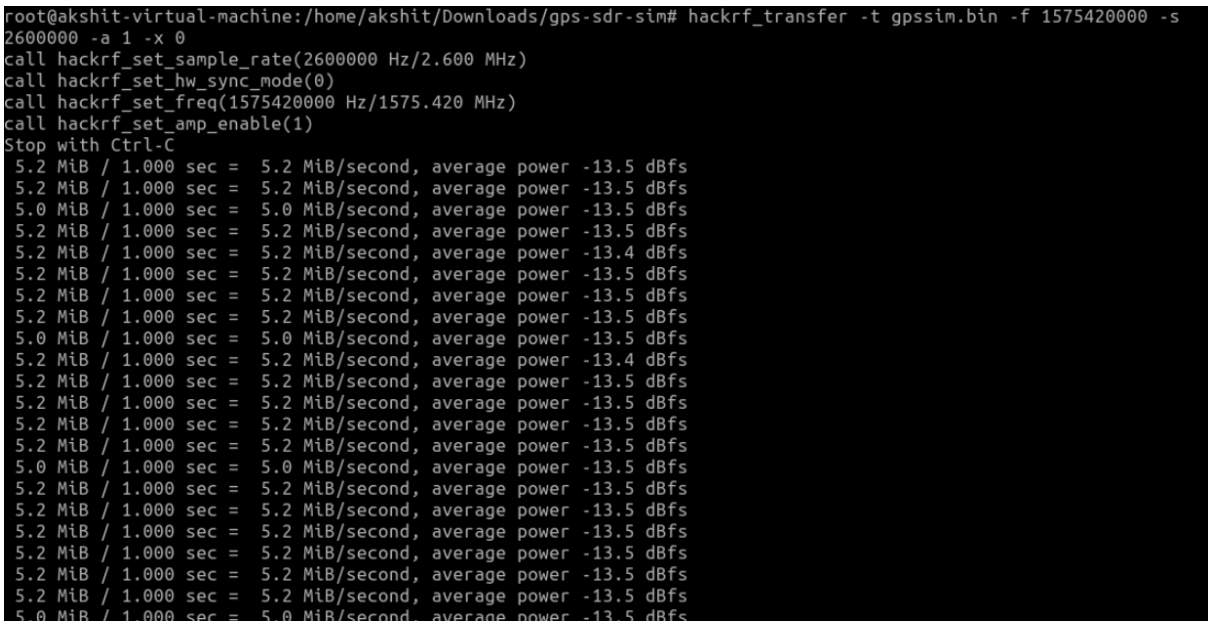
9. Transmitting the Spoofed GPS Signal

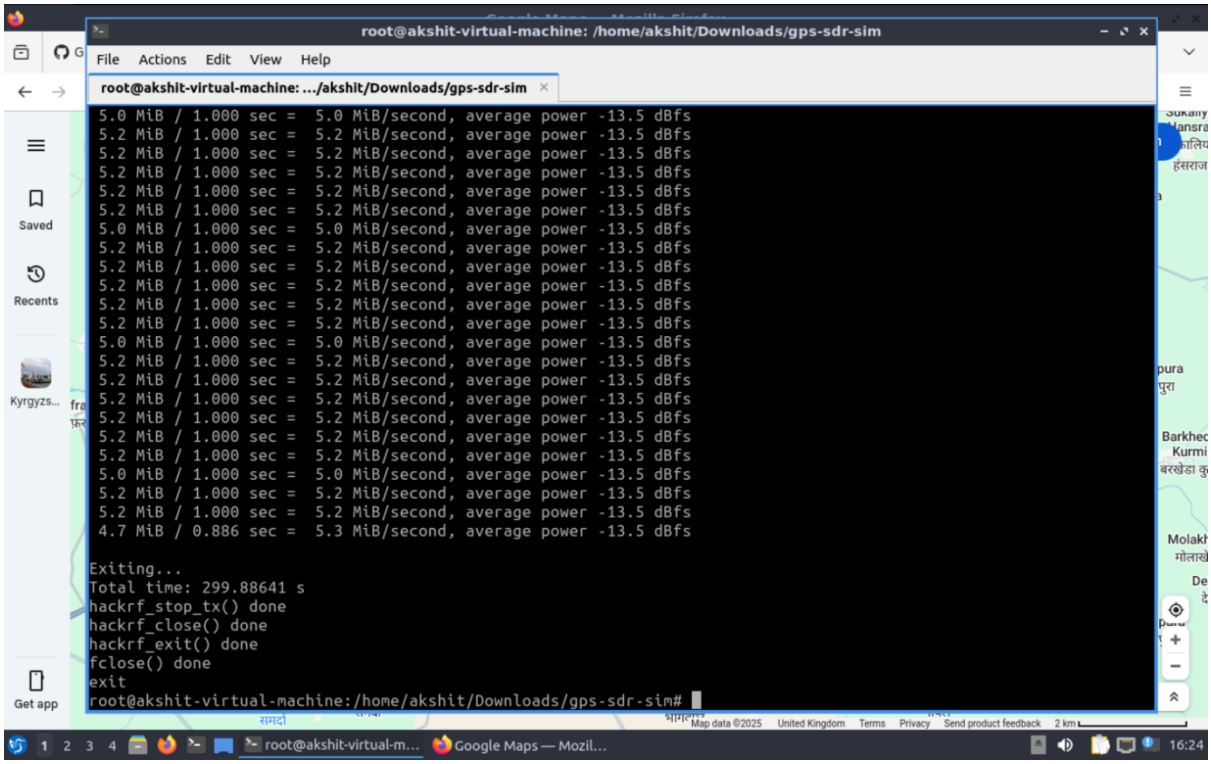
9.1 Transmission Command

hackrf_transfer -t gpssim.bin -f 1575420000 -s 2600000 -a 1 -x 0

Parameter explanation:

- -t gpssim.bin → The spoofed GPS waveform
- -f 1575420000 Hz → GPS L1 frequency (1575.42 MHz)
- -s 2600000 → Sample rate
- -a 1 → Enable amplifier
- -x 0 → TX gain (set to 0 for safety)





In a real environment, a GPS antenna will radiate these signals, which can overpower authentic GPS signals nearby.
Here, this was done only inside an RF-isolated environment.


GPS Antenna
Frequency:1575.42MHz
Voltage:3V~5V
SGS FC CE
Search here
Restaurants
Petrol
Attractions
Indira Gandhi
Int'l Airport
Hotel Pride Plaza
Top rated
Radisson Blu Plaza
Hotel, Delhi Airport
SAMALKA
Google Maps
New Delhi
23°
NAQI
Explore
You
Contribute
1575 420 000
0 dB
1.75 MHz
EXT

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10. Aviation Security Implications

This demonstration highlights critical aviation concerns:

✓ Aircraft rely heavily on GNSS

Spoofed GPS signals can mislead:

- Aircraft navigation systems
- Flight management systems
- ADS-B receivers
- Ground control navigation aids

✓ Growing real-world incidents

Recent GPS spoofing has been reported in:

- Middle East
- Mediterranean
- India (Delhi, Ladakh region)
- China
- Russia/Ukraine region

✓ Need for better protections

Airports and aviation agencies must:

- Deploy GNSS interference detection
- Use multi-constellation GNSS
- Add receiver authentication / RAIM+
- Rely on redundant navigation systems (INS, VOR/DME)

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11. Conclusion

This research project demonstrates how GPS spoofing works at a technical level and why modern aviation systems must implement robust countermeasures.
Using accessible tools like HackRF and GPS-SDR-SIM, it becomes clear that GPS spoofing is no longer theoretical — it is a practical threat.

Aviation authorities, cybersecurity professionals, and GNSS engineers must work together to enhance GPS integrity and ensure safe navigation for all aircraft.

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12. Real-Time Aircraft Tracking Using SDR: ADS-B and Mode S Decoding

To complement the GPS spoofing research, a real-time aircraft tracking system was also developed using Software-Defined Radio (SDR). This system receives, decodes, and analyzes ADS-B (Automatic Dependent Surveillance–Broadcast) and Mode S transponder signals transmitted by aircraft at 1090 MHz. The goal of this project is to understand how aviation telemetry works — and how GPS spoofing can influence ADS-B data downstream.

12.1 Project Description

A real-time aircraft tracking system was built using HackRF One, DragonOS, and SDR Angel to monitor aircraft positions, velocities, altitudes, ICAO identifiers, and other telemetry broadcast by aircraft within listening range.

This system demonstrates how aircraft navigation and surveillance heavily depend on GNSS correctness — making it directly relevant to GPS spoofing research.

12.2 Steps and Achievements

1. Hardware Setup

- Used HackRF One with PortaPack H2+
- Operated inside a DragonOS environment with SDR tools preinstalled
- Attached a 1090 MHz antenna for optimal aircraft signal reception

2. Software Configuration

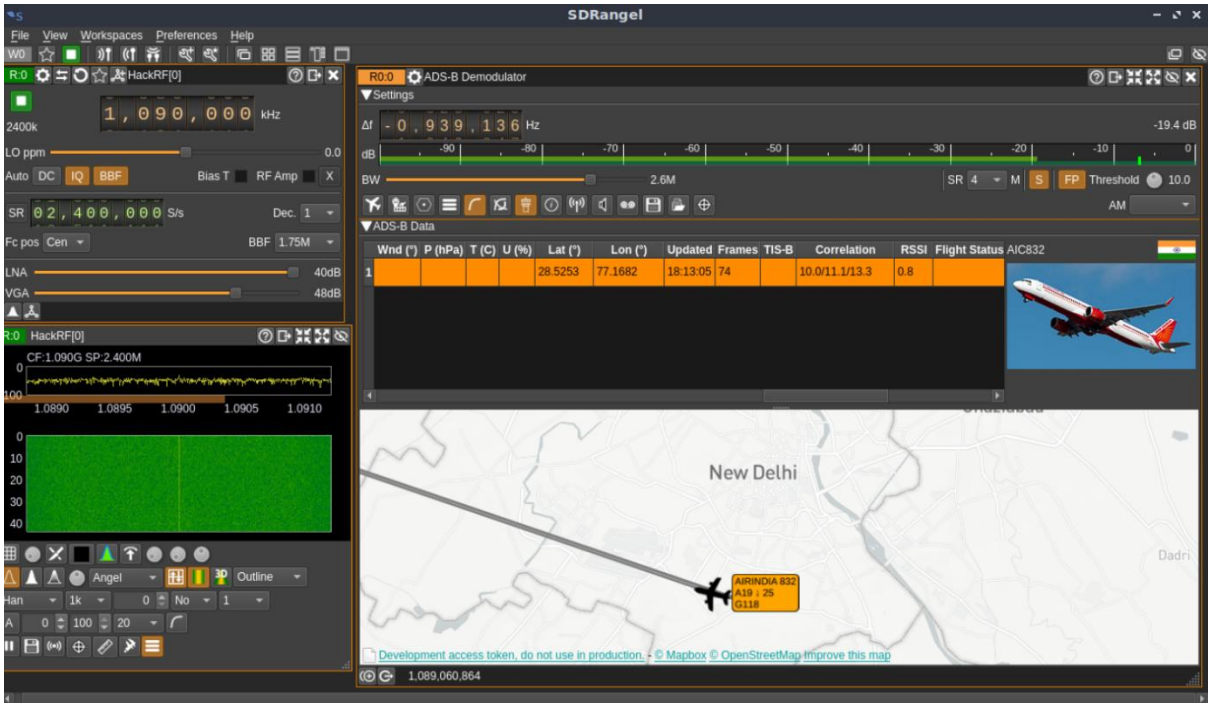
- Configured SDR Angel to:
    - Tune to 1090 MHz
    - Receive ADS-B and Mode S transmissions
    - Set appropriate sample rate for PPM demodulation

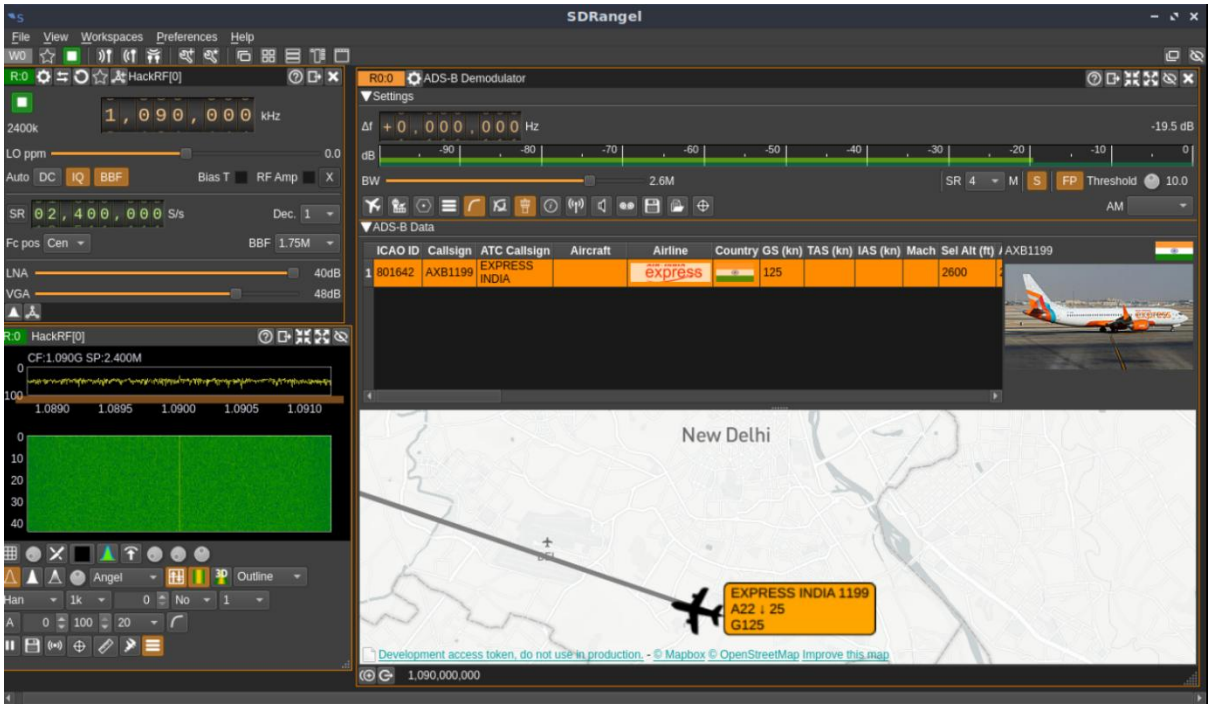
3. Signal Optimization

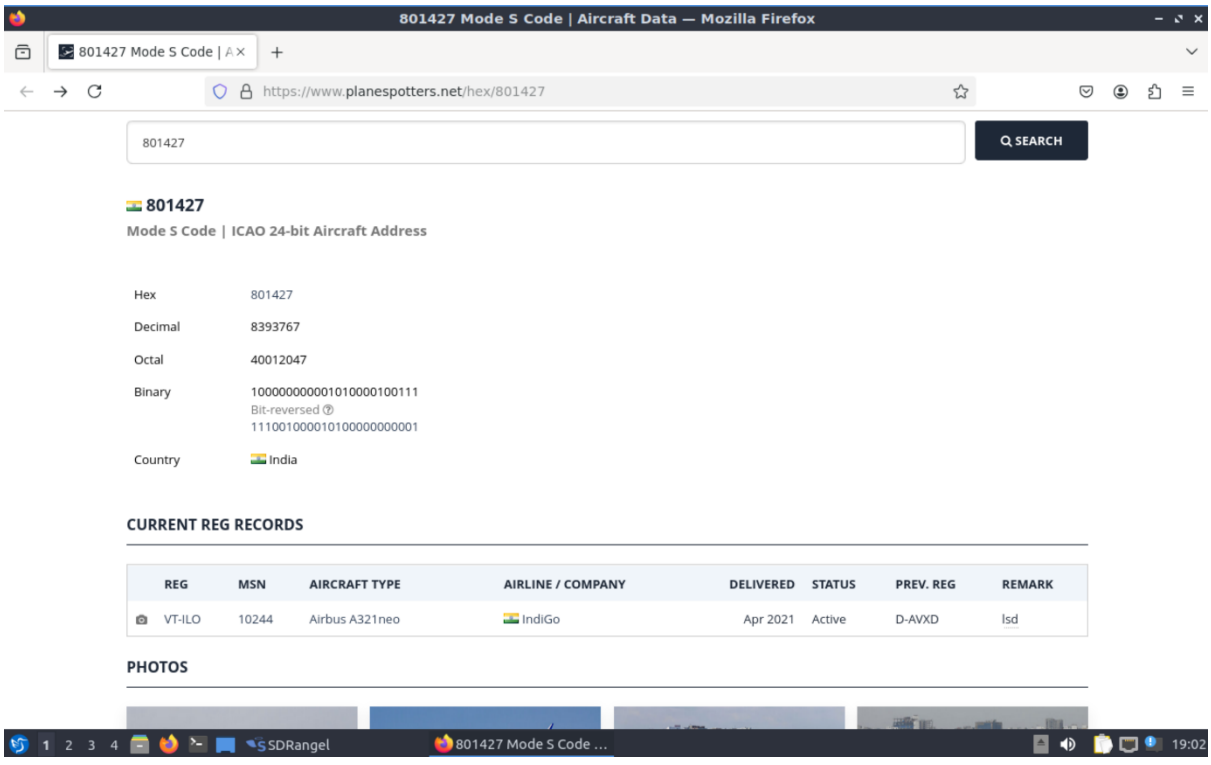
- Adjusted:
    - LNA (Low Noise Amplifier)
    - VGA (Variable Gain Amplifier)
      to maximize signal clarity
- Ensured minimal noise for clean decoding

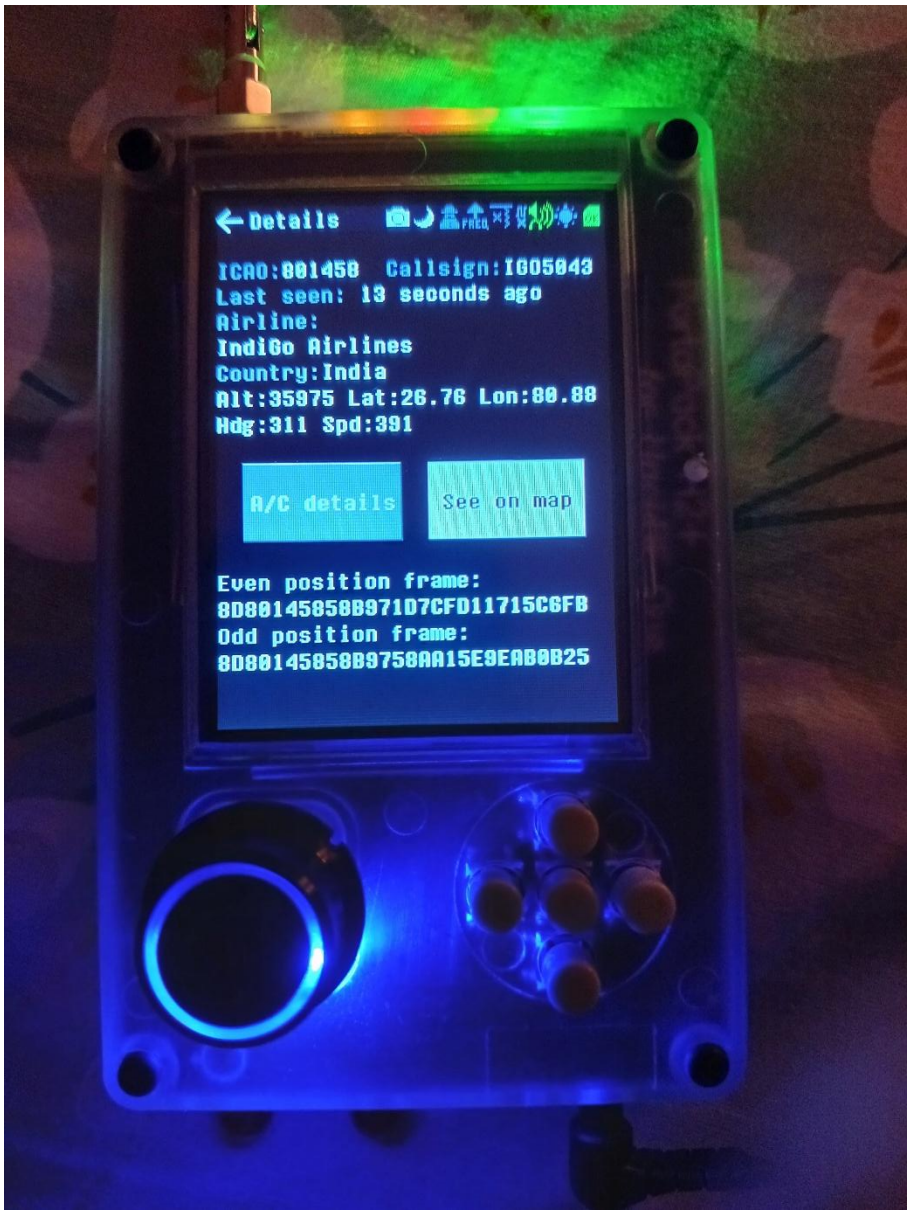
4. Demodulation and Decoding

- Added ADS-B Demodulator Channel in SDR Angel
- Successfully decoded:
    - Aircraft latitude & longitude
    - Altitude
    - Velocity
    - Heading
    - Squawk code
- Decoded Mode S 24-bit ICAO addresses to uniquely identify aircraft




SDRangel
File View Workspaces Preferences Help
HackRF[0]
1,090,000 kHz
ADS-B Demodulator
Settings
ADS-B Data
ICAO ID Callsign ATC Callsign Aircraft Airline Country GS (kn) TAS (kn) IAS (kn) Mach Sel Alt (ft)
801642 AXB1199 EXPRESS INDIA 125 2600
AXB1199
New Delhi
EXPRESS INDIA 1199
A22 ↓ 25
G125
Development access token, do not use in production. - © Mapbox © OpenStreetMap Improve this map
1,090,000,000


801427 Mode S Code | Aircraft Data — Mozilla Firefox
https://www.planespotters.net/hex/801427
801427
SEARCH
801427
Mode S Code | ICAO 24-bit Aircraft Address
Hex 801427
Decimal 8393767
Octal 40012047
Binary 100000000001010000100111
Bit-reversed
11100100001010000000001
Country India
CURRENT REG RECORDS
REG MSN AIRCRAFT TYPE AIRLINE / COMPANY DELIVERED STATUS PREV. REG REMARK
VT-ILO 10244 Airbus A321neo IndiGo Apr 2021 Active D-AVXD lsd
PHOTOS



5. Visualization

- Real-time aircraft data was displayed in SDR Angel, showing:
    - Aircraft telemetry
    - Flight movement
    - Transponder identity
    - Signal quality

12.3 Theory and Concepts

ADS-B (Automatic Dependent Surveillance–Broadcast)

- Aircraft broadcast their GPS-derived position multiple times per second

- Uses Pulse-Position Modulation (PPM)
- Broadcasts:
    - Position
    - Velocity
    - Identification
    - Navigation status

Mode S Transponders

- Provide aircraft identity (24-bit ICAO code)
- Used by ATC for tracking and interrogation

PPM Demodulation

- ADS-B messages encode bits based on the position of pulses
- Requires precise timing and signal clarity

SDR (Software-Defined Radio)

- Allows flexible demodulation and real-time processing
- Capable of decoding aviation telemetry without specialized hardware

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12.4 Skills Demonstrated

- SDR Signal Processing
- ADS-B & Mode S Decoding
- RF Gain Optimization
- GNSS-Based Telemetry Interpretation
- SDR Angel Configuration
- HackRF One Operations
- Aviation Cybersecurity Analysis

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13. Combined Attack Surface: Injecting GPS-Spoofed Coordinates into ADS-B Stream

With both systems operational —

1. GPS Spoofing with HackRF One, and
2. ADS-B/Mode S Real-Time Aircraft Tracking —

this research explored how spoofed GPS data can influence or mislead aviation surveillance systems by propagating into the ADS-B telemetry chain.

13.1 How GPS Spoofing Affects ADS-B

ADS-B depends on GPS-derived position.
If a GPS receiver receives spoofed GNSS signals, then:

✓ The aircraft's internal system believes the spoofed GPS coordinates

✓ The aircraft then broadcasts *those same fake coordinates* via ADS-B

✓ ATC systems, flight trackers, and SDR receivers see incorrect aircraft locations

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13.2 Combined Experiment Workflow

Step 1 — Generate Spoofed GPS I/Q Data

"./gps-sdr-sim -e brdc3190.25n -l 28.556276,77.100662,100 -b 8"

Step 2 — Transmit GPS Spoofed Signal

"hackrf_transfer -t gpssim.bin -f 1575420000 -s 2600000 -a 1 -x 0"

Step 3 — ADS-B Receiver Captures Modified Telemetry

Inside the RF test container:

- The GPS receiver locks onto the spoofed signal
- ADS-B/Mode S signals generated based on spoofed coordinates now reflect wrong locations

Step 4 — Real-Time Spoofed Location on SDR Angel

ADS-B visualization begins showing:

- Incorrect aircraft positions
- Incorrect velocities/headings
- Phantom movement patterns
- GNSS-induced telemetry errors

13.3 Findings

The combined system demonstrated:

✓ A spoofed GPS signal directly corrupts ADS-B positional data

✓ ADS-B has no authentication or GNSS integrity checks

✓ False aircraft locations can appear on tracking systems

✓ Airspace situational awareness can be misled

This highlights a critical vulnerability in aviation cybersecurity —
GNSS spoofing is not isolated to navigation; it cascades into surveillance, ATC workflows, and tracking systems.